The Higman–Sims incidence geometries: classification, constructions and corrected source data

Human Compression project

Abstract

We assemble complete finite proofs for the classification and two constructions in D. Leemans, *The residually weakly primitive geometries of HS*, together with the acquired supplementary diagrams. The classification counts are recovered. The diagrams, local groups and full automorphism groups are compared occurrence by occurrence, with corrections stated explicitly. Earlier sporadic classifications quoted by the article retain their own equivalence conventions and premises. The accompanying proof volume preserves the complete accepted arguments and finite certificates on which this synthesis depends.

1 The source and the mathematical statement

The fixed article is the six-page edition in Australasian Journal of Combinatorics **33** (2005), 231–236, denoted by L091. Its acquired supplement is the twenty-three-page edition dated 6 October 2003. The original results and constructions remain attributed to that article and its cited authors. Here an automorphism preserves each type; a correlation may permute the types. Isomorphism of geometries may also permute types. This convention is essential when counting the objects.

Let $G = \mathrm{HS}$ in the retained faithful permutation model on one hundred points, with $|G| = 44\,352\,000$. An eligible geometry is firm, residually connected and flag-transitive under G, residually weakly primitive, locally doubly transitive, and satisfies the geometric intersection condition $(\mathrm{IP})_2$. Weak primitivity means that at least one type in every nonempty residue is primitive. It does not assert primitivity on every type. The geometric condition $(\mathrm{IP})_2$ requires each rank-two residue to be a partial linear space or a generalized digon; it is distinct from an identity between subgroup intersections.

Theorem 1. *Under the explicit group and software inputs in Section 9, the eligible geometries have, up to incidence isomorphism, the following rank counts:*

rank	1	2	3	4	5	6	≥ 7
all geometries	1	6	30	41	8	1	0
not truncations	0	4	20	41	8	1	0.

The eighty-six classes of rank at least two correspond to the eighty-six supplementary diagrams after the explicit corrections recorded below. Their local data, full groups, thirteen printed truncation links, conjugacy notes and three residual-primitivity annotations have complete finite certificates. Five printed doubling links are verified; three are incompatible with the precise two-point operation used by the source adapter. Both Hoffman–Singleton constructions in the article are realized by explicit incidence isomorphisms.

The theorem concerns a corrected source disposition. It does not say that every printed expression is correct. In particular the original classification counts survive, while several subsidiary diagrams and group labels change. The separate comparison with other sporadic groups is stated in Section 8; its borrowed classification premises are not assertions proved by the HS census.

2 Coset recognition and complete enumeration

We recall the common argument because it identifies exactly what the finite enumeration proves. Suppose a group acts on the right, type-preservingly and flag-transitively, and choose a base chamber $(x_i)_{i\in I}$. Set $H_i = G_{x_i}$. The map

$$H_i g \longmapsto x_i^g$$

is a bijection on each type: two images agree precisely when their right cosets agree. Cosets meeting at g give incident translates of base objects. Conversely, flag transitivity on each pair of types gives a common representative for every incident pair. Thus this map preserves incidence and nonincidence.

For a chamber stabilizer B, let P_i be its panel stabilizers and put $K_\varnothing = B$, $K_S = \langle P_i : i \in S\rangle$. The recognition conditions are

$$K_I = G, \qquad K_S \cap K_T = K_{S\cap T}, \qquad [P_i : B] \geq 2, \tag{1}$$

together with typed coset Helly for the maximal parabolics $H_i = K_{I\setminus\{i\}}$. Typed Helly means that any pairwise intersecting family of cosets of distinct types has a common element. It is not inferred from the subgroup identity in (1).

Here is the exact finite Helly criterion used in the common proof. For every i, $J \subseteq I \setminus \{i\}$ with $|J| \geq 2$, and chosen $j_0 \in J$, require that each $g \in H_{j_0}$ satisfying $g \in H_i H_j$ for all $j \in J$ admit $k \in \bigcap_{j\in J} H_j$ with $H_i g = H_i k$. Necessity follows by applying Helly to $H_i g$ and the base cosets H_j. Conversely, induct on the number of cosets in a pairwise intersecting family. Translate a common element of all but one to the identity, then choose a representative of the remaining coset in H_{j_0}. The criterion supplies a common representative for the whole family, completing the induction.

These conditions recognize the desired firm, connected coset geometry. Helly gives flag transitivity and identifies residues with the coset systems in K_S. Equation (1) identifies all base flag stabilizers. Panels have at least two objects. The subgroup generated by the parabolics of a residue has exactly the objects reachable by coset paths; their generation of K_S proves connectedness. Conversely, residual connectivity reconstructs every complementary flag stabilizer from its panels, by induction on $|S|$, giving (1). The remaining tests are literal local conditions: maximal proper subgroups for primitivity, two-transitive panel actions, and common neighbours for $(\mathrm{IP})_2$.

Lemma 2. *Given complete representatives of the subgroup classes of G and of the maximal subgroups inside each representative, maximal augmentation enumerates every eligible panel family.*

Proof. If a maximal inclusion $M < K$ is transported so that $M^c = H$, every maximal overgroup of H in that inclusion class is $(K^c)^n$ for some $n \in N_G(H)$. Indeed, compare any transporter of the inclusion with the chosen c; their quotient normalizes H. Conversely every displayed conjugate is such an inclusion. This gives the complete actual list of panels above each fixed B, retaining only the required local two-transitive actions.

For a target of rank r, weak primitivity supplies a maximal parabolic H in its acting group K. Deleting the complementary panel leaves an eligible residue with the same Borel. By induction its panel tuple appears at the previous level. The missing panel lies in K and outside H, and the complete overgroup list includes K. Taking representatives under the stabilizer of the old tuple loses no extension: equivalent choices are related by an element fixing that tuple. The recognition and local tests accept exactly the eligible extensions, after which normalizer and type orbits remove duplicates.

Adjoining a panel strictly enlarges the generated group, since otherwise (1) would force that panel to equal B. Thus a tuple gives a strict subgroup chain. Its rank is bounded by $\Omega([G : B])$, counting prime factors with multiplicity. The finite record must reach an empty next level; a chosen rank cutoff alone is not an exhaustion certificate. Finally let B range over the complete subgroup catalogue. □

The accepted catalogue has 589 subgroup classes and 3075 maximal inclusion rows. The terminal inventory covers all 589 anchors: 529 have completed enumerations, and 60 have complete proper-generation certificates. For the latter, the join of every admissible panel is proper in G, so no selected family can generate G. More generally, for a partial family T, every possible added panel lies among those compatible with every member of T. If the join of this entire compatible set is proper, no full-group extension exists. Conjugation preserves compatibility, so rows may be transported along normalizer orbits. Every prefix of a genuine full-group family survives these bounds. Consequently these exclusions preserve the induction of Lemma 2; they are not invented empty search levels.

The complete input has 130 families under inner conjugation and type permutation, with rank counts $(6, 44, 67, 12, 1)$. Its full-automorphism quotient has counts $(6, 30, 41, 8, 1)$. To compare two tuples, fix a target type permutation, successively transport each subgroup, and replace the remaining acting group by the stabilizer of the subgroups already placed. The possible completions form precisely a transporter coset, so this procedure loses no solution. Every positive comparison retains its actual permutation and checks the entire family. The full ambient automorphism action is realized by the acquired permutation normalizer of order 88 704 000, with trivial centralizer of G.

3 Forgetting the specified action and removing truncations

The preceding quotient is initially a quotient of actions. An arbitrary incidence isomorphism need not preserve the specified G, so a further argument is required. For each type subset J, flag transitivity gives

$$\#\{\text{flags of type } J\} = |G : G_J|, \qquad G_J = \bigcap_{j \in J} H_j.$$

These are intrinsic incidence counts. Under a type permutation their coordinates are permuted. Since the group order is fixed, the stabilizer order profiles also separate incidence geometries. On the eighty-six representatives they separate all but pairs $(14, 15)$ and $(83, 84)$. The complete profile over all type subsets separates 83 and 84.

For $14, 15$, the distinct type sizes are $231000, 23100, 1100$. Define two objects of the last type to be adjacent when some object of the first type is incident with both. The resulting intrinsic graphs have 1100 vertices and valency 105. Blocks are determined by their sixteen-point shadows: the shadow stabilizer has exactly the block stabilizer's order 1920. Lines have repeated two-point shadows, so they are instead reconstructed from their four incident blocks. The stabilizer of that block set has order $4 \cdot 48 = 192$, exactly the line stabilizer. Restriction to points is therefore faithful.

Both complete graph groups have order 88 704 000, and their derived subgroup is the specified image of G. Any geometry isomorphism carries the intrinsic graphs and their derived subgroups to one another. By faithfulness this conjugation lifts to the full geometries. It would therefore induce an equivalence of the specified actions, already excluded by the complete family comparison. This proves that all eighty-six representatives remain distinct as incidence geometries. The exact graph upper bound is a retained algorithmic premise, not a consequence of checking a few positive automorphisms.

The complete rank-one panel inventory has two faithful two-transitive actions, both of degree 176. Rank-one incidence is just a set, so they give one bare class with full group Sym(176). Both have actual truncation maps to higher-rank representatives. There are no other rank-one actions satisfying the source conditions.

For ranks at least two, inspect every proper type subset of every representative. There are 756 subsets. Intrinsic type sizes, pair-flag counts and chamber counts leave exactly fifteen comparisons with a listed target. Every surviving comparison has an actual whole-family automorphism and type map, hence an incidence isomorphism on cosets. No negative assertion relies on a failed

unbounded isomorphism search. The targets are two distinct rank-two classes and ten distinct rank-three classes; none have higher rank. Since the parent list is complete and any isomorphism transports its type subsets, this tests every possible truncation. Subtraction proves the second row of Theorem 1. The thirteen printed supplementary links are obtained by composing these maps with the source coordinate maps.

4 Residues, local groups and source coordinates

For each class, the type sizes and node orders are respectively $|G : H_i|$ and $|G_{I\setminus\{i\}} : B| - 1$. The kernel of H_i on its whole residue is

$$K_i = \bigcap_{j \neq i} \mathrm{Core}_{H_i}(H_i \cap H_j).$$

For a rank-two residue with group H, parabolics P, Q and $D = P \cap Q$, the neighbours of Px are exactly Qux, where u runs through a right transversal of D in P. This formula constructs the actual incidence graph without confusing equality of counts with equality of incidences.

All 437 base rank-two residues are acquired, with 187005 vertices and 601049 edges in total. A breadth-first search from one vertex in each part determines the type diameters because the action is transitive on each part. It also determines the girth here. For a shortest cycle, transitivity places one vertex at the chosen root. The opposite vertex and its two neighbours have the distances prescribed by the shortest cycle; otherwise a shorter path would create a shorter cycle. One of the two opposite edges is a nontree edge, and its two root distances plus one give the cycle length. Conversely every such nontree value is at least the girth. Thus the minimum is exact. The diagram triple is the first diameter, half the girth, and the second diameter.

The source coordinates give a bijection from all eighty-six diagram labels to the proved classes. Most are fixed by the type-size multiset. The tied diagrams $3.22, 3.23$ are distinguished by their proved full groups, and $5.2, 5.3$ by their complete bond patterns. The explicit source-to-model type permutations are retained. This is identification within a complete classification, not a claim that unavailable historical generators equal the current permutations. Thirteen Petersen residues and two affine-plane residues have literal adjacency-preserving bijections to the two-subsets model of Petersen and to $\mathrm{AG}(2, 4)$ over $\mathbb{F}_2[t]/(t^2 + t + 1)$.

The local structural certificates identify all 86 Borels and 312 residue kernels, followed by all 312 full parabolics with their displayed kernels and quotients. The elementary transport argument is worth making explicit. Enumerate the complete Cayley graph of a named finite model M using its generators. Proposed images y_m in the actual subgroup must satisfy $y_1 = 1$ and $y_{ms} = y_m y_s$ on every generator edge. Induction on positive words proves multiplicativity. Distinct images, containment of the actual generators and the acquired order prove an isomorphism onto the actual subgroup. A discovery tree alone would not suffice: every remaining edge is also checked.

For a quotient by a normal subgroup K, the corresponding equations hold modulo K. The distinct products ky_m prove the exact index and quotient map. Exact edge equations give a section; commuting with K additionally gives a direct product. To disprove splitting for a fixed quotient, every ordered tuple of possible generator lifts is tested. In the three relevant S_5 quotients all 4096 lift tuples fail, proving nonsplitting. Merely failing to find a complement would not prove this. Dot notation is treated as an extension, not as an assertion of nonsplitting. Named models and their construction proofs, including the separate $\mathrm{Aut}(A_6)$ identification, remain explicit dependencies of these transports.

5 Full automorphism and correlation groups

For a transitive type with base stabilizer P, an equivariant encoding of its objects is injective exactly when the stabilizer of its base encoding equals P. An object may first be encoded by its

point shadow, then by the sets of already identified incident objects. Induction in the recorded dependency order proves that an automorphism fixing the chosen point type fixes every object. This is the common faithfulness argument; repeated shadows are not silently treated as distinct.

Several intrinsic relations give upper bounds on the point action. Let the point type have size n, fix a base point x incident with the type-j base object, and let that object's point shadow be S_j. Let O_k be an orbit of G_x of size d_k. The number of type-j objects incident with a pair in the corresponding ordered-pair orbit is

$$m_{jk} = \frac{[G : P_j]e_{jk}}{nd_k}, \qquad e_{jk} = |S_j|\,|S_j \cap O_k|.$$

The first equality counts object–pair incidences; the second follows from the transitivity of P_j on S_j. These formulas count objects even if shadows repeat. Sorting the data $(|G : P_j|, m_{jk})$ over types gives a relation intrinsic under all permitted type permutations. The complete graph group often has the specified G as its derived subgroup. Normality of G, an abelian quotient and $G' = G$ imply this equality. An arbitrary correlation then normalizes G, by characteristicity on points and faithful restriction to the geometry.

A second bound uses the link of a point in a family $\mathcal{B}$ of distinct point subsets. If A preserves $\mathcal{B}$ transitively on the point set X, its stabilizer at x embeds in the permutation group of $X \setminus \{x\}$ preserving $\{B \setminus \{x\} : x \in B \in \mathcal{B}\}$. Hence $|A| \le |X||L_x|$. If this equals the known subgroup order, equality determines the full typed group. Still other cases use an induced incidence system on several intrinsically selected types. Complete coloured graph bounds and literal lifting of their generators settle the two set constructions separately.

Once the specified G is normal in the full correlation group, put $B = \bigcap_i P_i$ and let S, T be the ordered and setwise family stabilizers in $\mathrm{Aut}(G)$. A chamber-fixing automorphism of G induces the exact coset map $P_i g \mapsto \alpha(P_i)\alpha(g)$. Conversely move a correlation's image chamber back by G and use its induced automorphism of G. This proves

$$|\,\mathrm{Aut}(\Gamma)| = |G : B||S|, \qquad |\,\mathrm{Cor}(\Gamma)| = |G : B||T|.$$

Every type permutation is tested by the complete transporter procedure. The centralizer of G in the correlation group is the actual quotient $(T \cap G)/B$. Its intersection with the typed group is trivial in the relevant cases. A checked central subgroup of order two, or a checked elementary subgroup of order four, and equality of product orders then prove the displayed direct products. They are not inferred solely from doubled group orders.

Eight accepted applications cover 82 representatives. Their disjoint sets have sizes 24, 9, 6, 6, 11, 5, 15 and 6. Cases 14, 15 and the constructions 35, 36 cover the remaining four. The full source table in the ledger contains all eighty-six names and their exact proof references. The following printed fields change; put $A = \mathrm{Aut}(\mathrm{HS})$.

Diagram	Field	Printed	Proved
3.20	Aut	G	A
3.20	Cor	A	$A \times C_2$
3.21	Aut	G	A
3.21	Cor	$G \times C_2$	$A \times C_2$
3.26	Aut	G	A
3.26	Cor	A	$A \times C_2$
4.22	Cor	$G \times C_2$	G
4.41	Cor	$A \times C_2$	A

The two blank fields of 3.10 are filled by A and $A \times C_2$. They are completed metadata, not two false printed groups.

6 The two Hoffman–Singleton constructions

In the literal model take 176 points, 1100 conics, each an eight-set, and 176 quadrics, each a fifty-set. The complete saved orbits verify the six Higman properties used in the article. Each point lies on fifty quadrics, and each conic on eight. Every point pair lies on two conics; the quadrics through that pair are exactly the union of those through the two conics, with precisely two quadrics containing both conics. Dually each quadric pair contains two conics, its point intersection is their union, and the conics meet in two points.

The sixth property is checked with both of its universal clauses. For a base conic C, form the 8 by 8 relation between its points and its incident quadrics: all conics meeting C in two points including the chosen point must lie in the chosen quadric; all conics lying in two quadrics through C, one the chosen quadric, must contain that point. Exactly one pair in each row and column satisfies both clauses. Transitivity on conics transports this matching to every conic. Literal point and quadric actions are doubly transitive, and the point stabilizer has quadric orbits of lengths $50, 126$, proving the additional transitivity assertions.

Let the lines be all $\binom{176}{2} = 15400$ point pairs. The first geometry has points, lines and conics, with containment incidence. The second has points, lines and quadrics. Point incidence is containment; a line meets a quadric exactly when the quadric contains both conics through that line. For each construction, the map from an actual coset $H_i g$ to the translate of its base subset is checked as a bijection. For every ordered pair of types, the subgroup orbit of a base object equals its entire geometric neighbour set. Equivariance proves that these maps preserve incidence and nonincidence everywhere. They identify the constructions with representatives $35, 36$, respectively, and with supplement diagrams $3.2, 3.1$.

Every line has two points and two planes; all four flags in its residue occur. Every incident point–plane pair has seven common lines. Thus all partial flags extend, and both geometries have 61600 chambers with Borel S_6. In the first geometry a point residue is a graph on fifty conics with 175 distinct edges, valency seven, diameter two and girth five; its conic residue is literally K_8. In the second, both point and quadric residues have the same fifty-vertex graph parameters. The classical uniqueness of the Hoffman–Singleton graph identifies these graphs. Their full group $U_3(5) : 2$ is retained as a named classical input; faithful actions of the actual parabolics attain its order. The line residues are digons. The line stabilizers have a literal S_6 kernel and two commuting involutions giving a disjoint complement of order four, proving $S_6 : 2^2$.

For the first construction the coloured point–conic graph has full group of order $|G|$; all its generators lift to the uniquely determined lines. Distinct type sizes exclude additional type permutations, so $\mathrm{Aut} = \mathrm{Cor} = G$. For the second, the marked point–quadric graph has full group of order $2|G|$. Its generators lift to lines using the verified bijection from point pairs to quadric pairs. The type kernel is G, and its centralizer is trivial, so conjugation identifies the full group with $\mathrm{Aut}(G)$. A saved type exchange supplies the duality. These constructions give the partial answer to the cited extension problem claimed by the original article; they are not newly solved open problems of this project.

7 Ancillary source statements and corrections

The 18 conjugacy notes involve 24 actual pairs of parabolics inside G: eleven positive transports and thirteen complete negative decisions. In 4.41 both pairs on the same level are conjugate and all four cross pairs are not. Abstract subgroup names alone do not decide these relations. The three stronger RPRI annotations, at $3.1, 3.2, 4.1$, require all inclusions $G_{J\cup\{i\}} < G_J$ to be maximal, not just one type per residue. All 56 inclusions are transported to actual rows in the complete maximal-inclusion catalogue. Finally each parenthetic source number is joined individually to its inner/type fusion fibre: 42 ones and 44 twos sum to 130. They are not additional incidence classes.

For doubling, the precise operation replaces a point type X and a type of distinct two-point

shadows by two copies X^+, X^-. An old edge joins the corresponding opposite copies; incidences with remaining types are inherited from the endpoints. When an edge is incident with another object exactly when both endpoints are, an old chamber gives a new chamber and conversely. The finite tests prove this condition, shadow injectivity and the whole-family target maps for

$$3.2 \to 3.3, \quad 3.4 \to 3.5, \quad 3.8 \to 3.9, \quad 4.8 \to 4.10, \quad 4.14 \to 4.19.$$

For $3.12 \to 3.15$ and $3.13 \to 3.16$, the only size-compatible choice has ten-point shadows. For $3.20 \to 3.21$, no choice can produce the new type size 30800. These are obstructions to the specified operation. The target geometries still exist, and no unspecified repair is claimed.

Six supplementary diagrams require numerical amendments. At 2.5 the relevant node order is 15, not 11; at 2.6 the type size is 44352, not 44320. The indicated triple in 3.15 is $(5, 4, 6)$ rather than $(6, 4, 5)$, and the top node order in 3.30 is 3, not 1. Diagram 4.22 has a digon at the lower bond and an omitted single diagonal bond between its 352-object types. In 4.28 the two indicated bonds have triples $(7, 5, 8)$ and $(4, 3, 3)$, the latter with the complete-graph incidence interpretation. Every correction is checked against all compatible type permutations in the complete class list.

The structural ledger retains eleven corrected occurrences of Borels or kernels and twenty-six parabolic occurrences. Five of the latter repeat the already corrected boxed kernels and are linked to those same occurrences; they are not counted again. The remaining parabolic clauses include three false splitting colons, ten affine quotients incorrectly printed as direct products, and the explicit quotient or full-group replacements listed in the ledger. Actual group maps prove the replacements; order, centre, element-order and complete-section arguments disprove the incompatible printed alternatives. A differently parenthesized but valid intermediate product in 4.12 is retained as correct.

Our own documentary mistakes are recorded separately. An old Table 1 transcription gave 153483 subgroups for J_1; the source actually prints 158485, agreeing with the proof. The intermediate product in 4.12 was once incorrectly read as a central direct factor of the full group. An auxiliary Janko transcription silently replaced a printed 17742 by the correct 17442; the source error and that transcription mistake are both retained. None of these is folded into a grand source-error count.

8 The comparison with the other sporadic groups

The article expressly quotes earlier classifications for the seven non-HS rows of Table 1. For a complete subgroup catalogue, the number of subgroups is $\sum_H [G : N_G(H)]$. In a complete table of marks its class lengths are $m_{H,1}/m_{H,H}$, because the two marks are $[G : H]$ and $[N_G(H) : H]$. Exact acquired rows give the following table. The rank columns remove only truncations of eligible higher-rank geometries in the same locally two-transitive class.

G	$\lvert G\rvert$	Classes	Subgroups	2	3	4	5	6
M_{11}	7920	39	8651	0	9	11	0	0
M_{12}	95040	147	214871	5	10	17	1	0
J_1	175560	40	158485	11	1	2	0	0
M_{22}	443520	156	941627	1	20	12	9	0
J_2	604800	146	1104344	5	16	3	0	0
M_{23}	10200960	204	17318406	0	3	16	10	0
HS	44352000	589	149985646	4	20	41	8	1
J_3	50232960	137	71564248	5	19	0	0	0

For the first six groups, the retained geometric equivalence is under $\mathrm{Aut}(G)$ and type permutations. No HS bare-isomorphism theorem is transferred to them without proof. The M_{12} rank-three value is 10 in that precise convention, compared with the printed 11. The independent HS argument above does prove its bare quotient.

The J_3 comparison has its own explicit boundary. Twenty-one acquired rank-three families have nineteen bare classes: intrinsic flag counts, two common-point-graph separations and IP/WIP separate them, while two actual whole-family maps identify the only repeated pairs. Its seven rank-two prescriptions are eligible and have distinct object-size pairs. All sixty-three type-pair profiles from the rank-three roster match exactly the two literal truncations already identified. This proves seven minus two equals five within that supplied population.

To interpret the last row globally, we retain from the article's reference [9] the earlier Janko classification's coverage and its absence of eligible geometries of rank at least four. The printed number seventeen is not imported as a premise: it counts diagram labels rather than the nineteen proved supplied classes. Reference [9] is historical L097, outside the fixed active 105 mathematical originals because its full article is unavailable; its acquired thirteen-page author supplement is auxiliary. No independent global Janko census or L097 recovery is claimed. The other auxiliary Janko metrics, names and full groups are not new assertions of L091 and are not additional conditions for its comparison-table disposition.

9 Attribution, premises and the proof volume

The following boundaries apply throughout.

1. The named faithful HS model, its complete subgroup catalogue, the complete maximal-inclusion lists, and the acquired full order and action of Aut(HS) are retained finite-group inputs. Their exact sources, generators, completed executions and review records are bound. The panel and maximal-deletion theory is reproduced here and in the common proof chapters; it is not replaced by trusting the target classification's printed totals.

2. Named small groups and extension models use their acquired construction or classification proofs. In particular the $\mathrm{Aut}(A_6)$ addendum is required in addition to the general parabolic transport certificates. Generic extension notation without its actual action is not treated as a unique group identifier. The addendum retains $A_6 \cong \mathrm{PSL}_2(9)$ and the full automorphism theorem identifying $\mathrm{Aut}(\mathrm{PSL}_2(9))$ with $\mathrm{P\Gamma L}_2(9)$. Conjugating automorphisms by the first isomorphism gives the required identification. The saved degree-ten semilinear model has exactly the same ordered generators as model 7003 in the earlier structural proof and model 319 in the HS library. The literal transports therefore land in that particular named group. Neither order 1440 nor the string $A_6 : 2 : 2$ supplies this identification by itself.

3. GAP's exact finite-group, subgroup, coset and transporter algorithms are software premises. Complete nauty or Traces upper bounds are also software premises. Literal positive generators, adjacency checks and small independent controls supplement these bounds; they do not alone prove completeness. No Lean verification is claimed by this intake.

4. The classical identification of the fifty-vertex Moore graph and its group $U_3(5) : 2$, and the historical Higman–Sims identification, retain their attribution to the cited work. The source's partial answer to the Hoffman–Singleton extension problem remains partial and is not counted as a new conjecture solved here.

5. The supplement attributes 3.4 to Ivanov–Shpectorov, 4.1 to Buekenhout's 1979 work, and 6.1 to the separate rank-six paper cited as reference [13] by the article. The present exact models recover their displayed occurrences but do not silently claim to reproduce those external constructions or complete another original.

The old mathematical results remain the work of the original authors. The present contributions are the common proof organization, the exact finite reproving and transports, the intrinsic incidence arguments that separate action equivalence from geometry isomorphism, and the explicit source corrections. These elementary arguments are not advertised as historically new solely because they appear in this project.

The complete proof volume preserves the accepted chapters unchanged. Their earlier local scope statements are historical; only the present occurrence ledger declares which later proof discharges each remaining clause. The acquisition manifest binds every included chapter and its full evidence chain. The final verifier checks that the source classes, local slots, group cases, annotations and correction links are complete and nonduplicated, then recompiles this synthesis. It does not repeat already accepted subgroup or graph searches. Original-level admission still requires a distinct review of this assembly. No compression factor or additional conjecture credit is assigned by joining these components.

10 Complete source class and group index

Here $G = \mathrm{HS}$ and $A = \mathrm{Aut}(\mathrm{HS})$. The numeric identifier fixes the literal family; the source-to-model type map and every local structural occurrence are in the accompanying ledger. An asterisk marks a corrected full-group field; a dagger marks the two completed blank fields.

Diagram	Page	Family	Multiplicity	Aut	Cor
2.1	3	6	1	A	A
2.2	3	5	1	A	A
2.3	3	4	1	A	A
2.4	3	3	1	G	A
2.5	3	2	1	A	A
2.6	3	1	1	A	A
3.1	4	36	1	G	A
3.2	4	35	2	G	G
3.3	4	34	2	G	$G \times C_2$
3.4	4	32	1	A	A
3.5	4	33	1	A	$A \times C_2$
3.6	5	31	1	A	A
3.7	5	29	1	A	A
3.8	5	30	1	A	A
3.9	5	28	1	A	$A \times C_2$
3.10	5	27	1	$A^\dagger$	$A \times C_2{}^\dagger$
3.11	6	26	1	A	$A \times C_2$
3.12	6	24	2	G	G
3.13	6	21	2	G	G
3.14	6	23	2	G	G
3.15	6	25	2	G	$G \times C_2$
3.16	7	22	2	G	$G \times C_2$
3.17	7	19	2	G	G
3.18	7	20	2	G	G
3.19	7	18	1	A	A
3.20	7	17	1	A^*	$A \times C_2{}^*$
3.21	8	16	1	A^*	$A \times C_2{}^*$
3.22	8	14	2	G	G
3.23	8	15	1	A	A
3.24	8	13	2	G	G
3.25	8	11	1	A	A
3.26	8	12	1	A^*	$A \times C_2{}^*$
3.27	9	10	1	G	A
3.28	9	9	2	G	G
3.29	9	8	2	G	G
3.30	9	7	2	G	$G \times C_2$
4.1	10	77	1	A	$A \times C_2$
4.2	10	76	1	A	A
4.3	10	74	1	A	A
4.4	10	75	1	A	A

Diagram	Page	Family	Multiplicity	Aut	Cor
4.5	11	73	1	A	$A \times C_2$
4.6	11	72	1	A	$A \times C_2$
4.7	11	71	1	A	$A \times C_2$
4.8	11	70	2	G	G
4.9	11	68	2	G	G
4.10	12	69	2	G	$G \times C_2$
4.11	12	67	1	G	A
4.12	12	66	2	G	G
4.13	12	60	1	A	$A \times C_2$
4.14	12	54	2	G	G
4.15	13	55	2	G	$G \times C_2$
4.16	13	59	2	G	$G \times C_2$
4.17	13	58	2	G	G
4.18	14	56	2	G	$G \times C_2$
4.19	14	57	2	G	$G \times C_2$
4.20	14	65	2	G	G
4.21	14	64	2	G	G
4.22	15	63	2	G	G^*
4.23	15	62	2	G	G
4.24	15	61	2	G	G
4.25	15	53	2	G	$G \times C_2$
4.26	16	52	2	G	$G \times C_2^2$
4.27	16	51	2	G	G
4.28	16	38	2	G	G
4.29	16	39	2	G	G
4.30	17	40	1	A	A
4.31	17	37	2	G	$G \times C_2$
4.32	17	49	1	A	A
4.33	17	50	1	A	A
4.34	18	48	2	G	G
4.35	18	47	2	G	G
4.36	18	43	2	G	G
4.37	18	44	2	G	G
4.38	19	41	2	G	G
4.39	19	45	1	G	A
4.40	19	42	1	G	A
4.41	20	46	1	G	A^*
5.1	20	85	1	A	$A \times C_2$
5.2	20	83	2	G	$G \times C_2$
5.3	21	84	2	G	$G \times C_2$
5.4	21	81	1	A	$A \times C_2$
5.5	21	82	1	A	$A \times C_2$
5.6	21	78	1	G	A
5.7	22	80	2	G	G
5.8	22	79	2	G	G
6.1	22	86	1	A	$A \times C_2$

11 Index to the complete proof volume

The companion volume appends the following accepted chapters without changing their mathematical text. The exact author and review records, including role disclosures, are in the acquisition manifest.

Chapter	Argument
01	Common action and panel theory

Chapter	Argument
02	Generation by all admissible panels
03	Compatible-panel generation
04	Transport of compatible panel sets
05	Fusion under full group automorphisms
06	Bare incidence separation and rank one
07	All higher-rank truncations
08	All local numerical data
09	Source coordinates and named residues
10	Elementary Borel and kernel structures
11	Compound Borel and kernel structures
12	All distinguished parabolic structures
13	Groups from one hundred points
14	Nine groups from joint point actions
15	Six groups from 352 points
16	Six groups from point links
17	Eleven groups from common incidences
18	Five groups from small incidence systems
19	Fifteen groups from larger point actions
20	The last six geometry groups
21	The two Hoffman–Singleton constructions
22	Five doublings and three qualifications
23	Conjugacy, primitivity and multiplicity annotations
24	The first seven comparison rows
25	The nineteen supplied Janko classes
26	The final Janko comparison join